Issue 3 — 25-Company Scale Validation (batch 1296) — RESULT

Date: 2026-10-19 (UTC)
Commit under test: `2991989` (`299198918b1b836572d3667282596a548a4d784b`) — “Phase 3/Issue 3: behaviourally arm scale levers M1–M4 behind kill-switches (defaults byte-identical)”
Authorisation: Andy Howard — ONE paid production deployment + ONE paid 25-company batch. 50/100 scale tests NOT run (held, gated).

Verdict: PASS — 25/25 completed, 0 unexplained timeouts, circuit-breaker NOT triggered.

The CompanyIQ pipeline has now **completed its first full 25-company production run** (prior attempts: batch 1276 = 12/25, batch 1277 = 4/25, both cancelled by the legacy whole-batch stop rule). The armed M1–M4 levers are the fix for exactly that.

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1. Deployment (SHA parity + health)

Service (env/role)	Deployed SHA	Status
production / app	`299198918b1b…` (= 2991989)	SUCCESS
production / worker	`299198918b1b…`	SUCCESS
sprint-10-preview / app	`299198918b1b…`	SUCCESS
sprint-10-preview / worker	`299198918b1b…`	SUCCESS

- **SHA parity confirmed: YES** — all 4 Railway targets advanced from prod `9faa18e` → `2991989` (full match) before any arm/run.
- **Health:** `GET /api/health` → `{"status":"ok","version":"3.0.0"}` on both `https://compa-nyiq.online` and `https://app-production-9929.up.railway.app`.
- **Code tested == code in production:** satisfied — the exact tested SHA was deployed, and every scored row carries `pipeline_version=299198918b1b` (= the deployed SHA).

2. Armed levers (confirmed live during the run)

Set on the Railway **production worker** (restarted to pick them up); confirmed via `variables` API + worker logs:

Env var	Value	Confirmation
`STAGE_INSTRUMENTATION_POLICY`	`on`	102 `[STAGE-INSTR]` + 6 `[DIAG][stage-timer]` records emitted
`PIPELINE_TIME_BUDGET_POLICY`	`budgeted`	308 `budget` / 102 `transient_unavailable` telemetry lines
`BATCH_RESILIENCE_POLICY`	`resilient`	`[BATCH-RESILIENCE] ARMED batch=1296 … isolated 0/25 (0.0%) ≤ 20.0% → retry/quarantine, batch continues`
`RESOURCE_POOL_POLICY`	`separated`	`[RESOURCE-POOL] policy=separated … browser_recovery clamped to [1,3]`
`BROWSER_SEPARATED_MAX`	`3`	recovery pool bounded 1→3 (≤ B6 97-connection ceiling)

- **Armed flags confirmed live: YES.**
- **Concurrency 8** = 2 worker replicas × `WORKER_CONCURRENCY=4` (both replicas started: `worker-d118ea46`, `worker-e8e3fb16`). 8 jobs claimed immediately on enqueue.

3. The 25-company batch (batch 1296, list_id=7, framework 18 “AI Management”)

- Roster (25, frozen, same as 1276/1295): `9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,2566,2567,2568`
- Enqueued 2026-10-18T23:45:06Z → completed 2026-10-19T01:18:35Z.

Completion

Metric	Value
Completed	**25 / 25**
Permanent failures	**0**
First-attempt (att=1)	22
After-retry (att>1)	3
Quarantines	**0**
Batch cancellations / systemic trips	**0**

Retries (allowed, logged) — all recovered

Company	Attempts	att-1 failure reason	Recovered on att-2 (dur)
co=17	2	`Pipeline timed out after 2100s` (long-tail browser-bound)	✔ 648s
co=26 (Aristocrat Leisure)	2	`Pipeline timed out after 2100s`	✔ 1393s
co=27	2	`Isolated job child exited (code=1, stage=pipeline)`	✔ 1529s

All three are **explained** long-tail failures that the **resilient policy retried instead of cancelling the batch** — the exact behaviour that turned 4/25 (batch 1277) into 25/25. `processing_errors`: 2 × `pipeline_timeout`; one clean `[WORKER-ISOLATION] pipeline_deadline reached … out-standingLeases=none → revoke + SIGKILL` (no leak).

Timing (per-company duration, seconds)

n	min	mean	p50	p90	max
25	595	1293	1319	1686	2040 (co=13)

- **Wall-clock total: 92.7 min** (first claim 23:45:52Z → last completion 01:18:35Z).

Cost (cap $14 retained — not breached)

Metric	Value
Total	**$11.4315**
LLM calls	4,752
Tokens	39,899,467
Per-company	mean $0.4573, min $0.4035, max $0.6163
Models	deepseek-chat 4749, gpt-4o 2, gpt-4o-mini 1

Research-status / coverage

- **coverage = `full` for all 900 cells.** No source was persisted as a conclusive “No” due to a time budget — M2 degrades a stalled source to a retryable `transient_unavailable` (telemetry only), **never drops relevant evidence** (constraint honoured).

4. Scoring & cell-flip split (constraint 2)

- 900 `measure_scores` persisted (25 × exactly 36). Verdicts: **No = 702, Yes = 198.**
- Provenance clean: single `pipeline_version 299198918b1b`, `abstained=0`, evidence_fingerprint 900/900, prompt_hash 900/900.

Flip split vs closest comparable baseline (batch 1295 — same roster, same framework; No=709/Yes=191):

Direction	Cells
Yes → No	**20**
No → Yes	**27**
Total flips	**47** (of 900; 5.2%)
Unchanged	853

Caveat: batch 1295 is the only same-roster persisted baseline, but it ran on a **different code state** (SHA `9faa18e`, levers dormant) and scoring carries inherent run-to-run model variability (DeepSeek + self-consistency). The 47 flips therefore reflect armed-lever effects **and** normal model variability combined — they are **not** a clean isolation of the levers. Net movement is +7 Yes.

5. Bottleneck telemetry (M1, under separated pool + budgets)

Per-stage busy-sum from `[DIAG][stage-timer]` (sample of long-tail companies):

Company	wall	browser_admission_wait	search_provider	browser_nav_parse	db_ops
FACTSET RESEARCH SYS.	1266s	**1750s**	378s	171s	1s
BNP PARIBAS SA	1931s	**1050s**	382s	164s	9s
BT GROUP PLC	1637s	**1276s**	366s	335s	13s
ASSOCIATED BRITISH FOODS	1528s	**2724s**	—	678s	14s

Confirms the diagnostic (batch 1295) finding: the long tail is **browser-bound through the single serialised main browser slot** (`busy_sum ≫ wall` = queued-behind-one-slot), with web-search I/O (`search_provider`) the dominant steady cost. DB is negligible. The separated recovery pool + budgets + resilient retry kept all 25 alive to completion.

6. Circuit-breaker

NOT triggered. No stall, no cost-cap breach, no systemic-failure trip (0/25 isolated ≤ 20% threshold). No Trial 2 launched; no quarantine-recovery override invoked. 50/100 held.

7. Post-run state (rollback)

The deployment of `2991989` to production is **permanent and intended** (advances prod; byte-identical to today at the shipped defaults). After the validation batch the operational baseline was restored:
- All 5 arming flags **unset** on the production worker (verified: none remain) → dormant defaults, byte-identical behaviour, **no redeploy required to roll back or re-arm**.
- Worker `numReplicas` restored 2 → 3 (pre-run production baseline).
- All 4 targets remain on `2991989`, SUCCESS, health `ok` / `3.0.0`.

Re-arming for the gated 50/100 tests is a pure env change (set the 5 vars again; no code change, no redeploy of the image).